

1. Administrative & Study Identification

Full Study Title: A Phase 2 Open-label Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of Sotatercept (MK-7962) in Children from 1 to Less Than 18 Years of Age With PAH on Standard of Care **Short Title/Acronym:** Study to Evaluate Sotatercept (MK-7962) in Children With Pulmonary Arterial Hypertension (PAH) (MK-7962-008) **Protocol Number:** MK-7962-008 **NCT Number:** NCT05587712 **Posting ID:** 925737704391448480 **Trial Status:** Recruiting **Sponsor/Company Name:** Merck **Investigational Product:** Sotatercept (MK-7962) **Phase of Development:** Phase 2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability, and pharmacokinetics (PK) of sotatercept over 24 weeks of treatment in children ≥ 1 to <18 years of age with PAH World Health Organization (WHO) Group 1 on standard of care (SoC). There is no formal hypothesis. **Secondary Objectives:** To evaluate the mean change from baseline in 6-Minute Walk Distance (6MWD), Tricuspid Annular Plane Systolic Excursion (TAPSE), and Pulmonary Artery Systolic Pressure (PASP).

2.2 Background & Rationale The safety and efficacy of sotatercept, a first-in-class activin signaling inhibitor, have not been well-studied in the pediatric population. This trial aims to address this gap by generating clinical data on the safety, tolerability, and pharmacokinetics of sotatercept in children with PAH WHO Group 1 who are already receiving standard-of-care treatments.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Participants are allocated to age-based dosing groups)

3.2 Planned Enrollment & Duration Target N\$: 42 participants **Number of Sites:** Multicenter **Estimated Study Start:** 19-Jan-2023 **Estimated Primary Completion:** 21-Sep-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female children ≥ 1 to <18 years of age. 2. Documented historic diagnostic right heart catheterization (RHC) any time before Screening confirming the diagnosis of PAH WHO Group 1 in any of the following subtypes: Idiopathic pulmonary arterial hypertension (IPAH), Heritable PAH, Drug/toxin-induced PAH, PAH associated with connective tissue disease, PAH-congenital heart disease (CHD) with shunt closure >6 months before Screening and subsequently confirmed by RHC before Screening, or PAH with coincidental shunt. 3. Must be on a stable dose(s) of background PAH therapy (phosphodiesterase-5 (PDE5) inhibitors, endothelin receptor antagonists (ERAs), soluble guanylate cyclase stimulators (sGCS), or prostanoids [including subcutaneous and intravenous]). 4. If male, agrees to abstain from heterosexual intercourse as their preferred and usual lifestyle or uses contraception during the intervention period and for at least 16 weeks (112 days) after the last dose of study intervention. Also agrees to refrain from donating blood or sperm for the duration of the study and for 16 weeks (112 days) after the last dose. 5. If female, must be either not a woman of childbearing potential (WOCBP) or use a contraceptive method that is highly effective or be abstinent from heterosexual intercourse during the intervention period and for at least 16 weeks (112 days) after the last dose. Also agrees to refrain from donating blood, eggs, or ovum for the duration of the study and for at least 16 weeks (112 days) after the last dose.

4.2 Exclusion Criteria 1. History of left-sided heart disease, including valvular disease (eg, moderate or greater mitral or aortic regurgitation or stenosis), left ventricular outflow tract obstruction, and/or left heart failure (eg, restrictive or dilated cardiomyopathy). 2. Severe (as based on the opinion of the investigator) congenital or developmental abnormalities of the lung, thorax, and/or diaphragm. 3. History of Eisenmenger syndrome, Potts shunt, or atrial septostomy. 4. Unrepaired or residual cardiac shunt. 5. Diagnosis of pulmonary veno-occlusive diseases, pulmonary capillary hemangiomatosis, or overt signs of capillary and/or venous involvement. 6. PAH associated with portal hypertension. 7. Known visceral (lung, liver, or brain) arteriovenous malformation(s). 8. History of full or partial pneumonectomy. 9. Untreated more than mild obstructive sleep apnea. 10. History of known pericardial constriction. 11. Family history of sudden cardiac death or long QT syndrome. 12. Any current or prior history of symptomatic coronary disease (myocardial infarction, percutaneous coronary intervention, coronary artery bypass graft surgery, or cardiac anginal chest pain) within 6 months before Screening. 13. Cerebrovascular accident within 3 months before Screening. 14. Prior exposure to sotatercept or luspatercept or has had an allergic reaction to any of their excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sotatercept (MK-7962) 0.3 mg/kg to 0.7 mg/kg administered once every 3 weeks. Dosing is stratified by age group: 12 to 18 years start at 0.3 mg/kg rising to 0.7 mg/kg; 1 to 12 years receive 0.3 mg/kg. **Route of Administration:** Injection. **Duration of Treatment:** Initial treatment period of 24 weeks (6 months), followed by an optional extension phase of up to 6 additional years.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background standard of care (SoC) therapies for PAH. **Prohibited:** Unapproved investigational agents or therapies outside of standard PAH protocols.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Historic RHC review
Baseline	Day 0	First Dose, Baseline Cardiac MRI, PK/PD Labs, Quality of Life Questionnaires
Interim	Every 3 weeks	Site visits for Treatment Administration, Safety Labs, Efficacy Assessment, PK Evaluation
End of Study	Week 24	Final 24-week assessment (including final Cardiac MRI), Transition to optional 6-year extension phase

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Serum Trough Concentration (C_{trough}) of Sotatercept:** C_{trough} was the lowest concentration of Sotatercept in serum just before the next dose. Blood samples will be collected at multiple time points to estimate the C_{trough} of Sotatercept.
- 2. Area Under the Curve at Steady State (AUC_{ss}) of Sotatercept:** Blood samples will be collected at multiple time points to estimate the AUC_{ss} of Sotatercept.
- 3. Area Under the Curve from 0 to 3 weeks (AUC_{0-3 weeks}) of Sotatercept:** Blood samples will be collected at Predose Day 1, Day 7, Day 14, and Predose Day 21 to estimate the AUC_{0-3 weeks} of Sotatercept.

7.2 Secondary Endpoints

- 1. Mean Change from Baseline in 6-Minute Walk Distance (6MWD) (Cohorts 1 and 2).**
- 2. Mean Change from Baseline in Tricuspid Annular Plane Systolic Excursion (TAPSE).**
- 3. Mean Change from Baseline in Pulmonary Artery Systolic Pressure (PASP).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any worsening of a preexisting condition or new untoward medical

occurrence temporally associated with the sponsor's product. Hematological parameters (e.g., hemoglobin) are also routinely assessed. **Serious Adverse Events (SAEs):** Reported per standard sponsor safety protocols. **Data Monitoring Committee (DMC):** Utilizes standard Merck / MSD internal and external safety monitoring committees.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE evaluations and PK analysis population for steady-state concentration assessments. **Sample Size Justification:** N=42 participants; this is an exploratory Phase 2 study with no formal hypothesis testing. **Handling of Missing Data:** Handled per standard sponsor Statistical Analysis Plan (SAP) using descriptive statistics and applicable imputation methods.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Periodic site visits, clinical laboratory assays evaluation, and blinded independent central review (BICR) for cardiac MRI interpretation. **Audit Trail:** Electronic Case Report Form (eCRF) tracking and standardized data validation.

11. Study Identifiers & Governance

Primary ID: MK-7962-008 **Posting ID:** 925737704391448480 **Registry Number:** NCT05587712 **Sponsor/Lead Organization:** Merck **Study Title:** Study to Evaluate Sotatercept (MK-7962) in Children With Pulmonary Arterial Hypertension (PAH) (MK-7962-008) **Official Regulatory Title:** A Phase 2 Open-label Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of Sotatercept (MK-7962) in Children From 1 to Less Than 18 Years of Age With PAH on Standard of Care

12. Clinical Framework (Adapted for PAH)

Primary Condition: Pulmonary Arterial Hypertension (PAH) **Disease Area:** Pulmonary Arterial Hypertension **Diagnosis Threshold:** Confirmed via historical right heart catheterization (RHC) **Medical Methodology:** Standard of Care (SoC) plus investigational activin signaling inhibitor **Optimization Target:** Hemodynamic improvement, symptom reduction, and pharmacokinetic profiling

13. Study Procedures & Methodology

Management Pathway: Evaluation of add-on therapy to standard PAH clinical pathway. **Education Protocol (HCP):** Site initiation and study drug administration training. **Education Protocol (Patient):** Informed consent (or assent), treatment counseling, and optional extension planning. **Quality Audit Mechanism:** Standard sponsor auditing for Phase 2 multicenter trials.

14. Observation & Follow-Up Schedule

Total Duration: 24 weeks initial treatment, plus an optional 6-year extension **Onsite Visit Cadence:** Every 3 weeks **Remote Monitoring Frequency:** N/A (Standard site visits apply) **Checklist Review Interval:** Every 3 weeks synchronized with dose administration

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					